

AI BlogNest API

Sprint Planning & Estimation Document

Date	15 February 2025
Team ID	SWTID-2026-6999
Project Name	AI BlogNest API
Maximum Marks	5 Marks
Project Duration	20 September 2026 - 26 September 2026

Team Details	
Team Leader	Ganesh Kumar T
Team Member	Hariharan R
Team Member	Dinesh Karthick V
Team Member	Santhosh M

1. Product Backlog, Sprint Schedule and Estimation

The AI BlogNest API development is planned using an Agile sprint-based approach. The project is divided into four sprints so that authentication, blog management, AI functionality, search, security, and testing can be developed in an organized sequence. Each backlog item includes the Sprint, Functional Requirement/Epic, User Story Number, User Story/Task, Story Points, Priority, and Team Members.

Sprint-1

Sprint	Functional Requirement (Epic)	USN	User Story / Task	Points	Priority	Team Members
Sprint-1	Authentication	USN-1	As a user, I can register for the application using my name, email and password.	3	High	R. Hariharan
Sprint-1	Authentication	USN-2	As a user, I can log into the application using my email and password.	2	High	R. Hariharan
Sprint-1	Authentication	USN-3	As a user, I can access protected APIs using JWT authentication.	3	High	R. Hariharan
Sprint-1	User Management	USN-4	As an administrator, I can manage users and their roles.	2	High	R. Hariharan
Sprint-1	RBAC	USN-5	As a system, I can restrict API access according to Admin, Editor, Author and Reader roles.	2	High	R. Hariharan
Sprint-1	Backend Setup	USN-6	As a developer, I can organize the backend using MVC architecture.	2	High	R. Hariharan
Sprint-1	Database	USN-7	As a system, I can connect and store application data using MongoDB Atlas.	3	High	R. Hariharan
Sprint-1	API Testing	USN-8	As a developer, I can test the authentication APIs using Postman.	3	Medium	R. Hariharan
Sprint-1 Total				20		

Sprint-2

Sprint	Functional Requirement (Epic)	USN	User Story / Task	Points	Priority	Team Members
Sprint-2	Blog Management	USN-9	As an Author, I can create a blog post.	3	High	R. Hariharan
Sprint-2	Blog Management	USN-10	As a user, I can view published blog posts.	2	High	R. Hariharan
Sprint-2	Blog Management	USN-11	As an Author, I can update my own blog posts.	2	High	R. Hariharan
Sprint-2	Blog Management	USN-12	As an authorized user, I can delete blog posts according to permissions.	2	Medium	R. Hariharan
Sprint-2	Blog Workflow	USN-13	As an Author/Editor/Admin, I can manage blog publishing states.	3	High	R. Hariharan
Sprint-2	Category Management	USN-14	As an authorized user, I can create and manage blog categories.	3	Medium	R. Hariharan
Sprint-2	Comment Management	USN-15	As a user, I can add comments to published blogs.	2	Medium	R. Hariharan
Sprint-2	Comment Moderation	USN-16	As an Editor/Admin, I can moderate comments.	3	Medium	R. Hariharan
Sprint-2 Total				20		

Sprint-3

Sprint	Functional Requirement (Epic)	USN	User Story / Task	Points	Priority	Team Members
Sprint-3	AI Content Generation	USN-17	As an Author, I can generate blog content from a topic using Google Gemini.	5	High	R. Hariharan
Sprint-3	AI Summarization	USN-18	As a user, I can generate a summary of blog content using Google Gemini.	3	High	R. Hariharan
Sprint-3	Prompt Engineering	USN-19	As a system, I can use structured prompts to generate consistent AI output.	3	High	R. Hariharan
Sprint-3	Keyword Search	USN-20	As a user, I can search blogs using keywords.	3	High	R. Hariharan
Sprint-3	Atlas Search	USN-21	As a system, I can use MongoDB Atlas Search for full-text blog searching.	3	High	R. Hariharan
Sprint-3	Embeddings	USN-22	As a system, I can generate and store vector embeddings for blog content.	3	High	R. Hariharan
Sprint-3 Total				20		

Sprint-4

Sprint	Functional Requirement (Epic)	USN	User Story / Task	Points	Priority	Team Members
Sprint-4	Vector Search	USN-23	As a user, I can search blogs using semantic meaning.	4	High	R. Hariharan
Sprint-4	Atlas Vector Search	USN-24	As a system, I can retrieve relevant blogs using MongoDB Atlas Vector Search.	4	High	R. Hariharan
Sprint-4	Security	USN-25	As a system, I can validate and sanitize incoming API requests.	3	High	R. Hariharan
Sprint-4	Security	USN-26	As a system, I can limit excessive API requests using rate limiting.	2	Medium	R. Hariharan
Sprint-4	Testing	USN-27	As a developer, I can test all major APIs through Postman and automated tests.	3	High	R. Hariharan
Sprint-4	Documentation	USN-28	As a developer, I can prepare technical documentation and project demonstration material.	2	Medium	R. Hariharan
Sprint-4	Final Integration	USN-29	As a system, I can integrate authentication, blog management, AI and search into one backend application.	2	High	R. Hariharan
Sprint-4 Total				20		

2. Sprint Summary

Sprint	Total Story Points	Duration	Planned Start Date	Planned End Date
Sprint-1	20	2 Days	20 Sep 2026	21 Sep 2026
Sprint-2	20	2 Days	22 Sep 2026	23 Sep 2026
Sprint-3	20	2 Days	24 Sep 2026	25 Sep 2026
Sprint-4	20	1 Day	26 Sep 2026	26 Sep 2026

Note: Only the overall project window (20–26 September 2026) was provided. The four sprints have been fitted sequentially into this 7-day span rather than the original 6-day/sprint template estimate. Adjust individual sprint boundaries if your course requires the literal 6-day durations.

3. Project Tracker

Sprint	Total Story Points	Duration	Start Date	End Date	Story Points Completed	Release Date
Sprint-1	20	2 Days	20 Sep 2026	21 Sep 2026	20	21 Sep 2026
Sprint-2	20	2 Days	22 Sep 2026	23 Sep 2026	20	23 Sep 2026
Sprint-3	20	2 Days	24 Sep 2026	25 Sep 2026	20	25 Sep 2026
Sprint-4	20	1 Day	26 Sep 2026	26 Sep 2026	20	26 Sep 2026

4. Velocity

Velocity represents the number of story points planned or completed during a sprint.

Total planned story points = 80 | Number of sprints = 4
Average planned velocity = $80 \div 4 =$ **20 story points per sprint**

20

STORY POINTS PER SPRINT — AVERAGE VELOCITY

5. Burndown Chart Data

The burndown chart represents the amount of work remaining against the progress of the project — a graphical representation of work left to complete versus time.

Sprint	Planned Story Points Remaining
Start	80
After Sprint-1	60
After Sprint-2	40
After Sprint-3	20
After Sprint-4	0

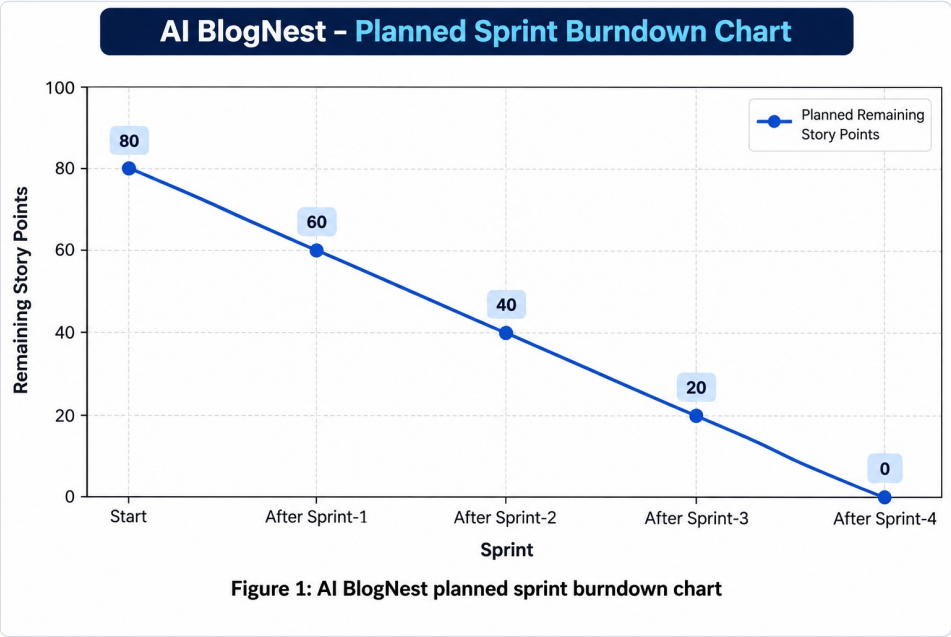


Figure 1: AI BlogNest planned sprint burndown